

Voltage sag source location: A review with introduction of a new method

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ABSTRACT

In this paper different methods for voltage sag source location (upstream or downstream) based on various criteria such as energy, impedance, voltage or current will be simulated and compared with each other. Then, a new method based on directional overcurrent (DOC) relay information or by applying its algorithm will be introduced. In this method the current (presag and during sag) is the only variable measured by relay/power quality (PQ) monitor. Then, the voltage sag source relative location is found using variation of positive-sequence magnitude of the measured current and the sign of its phase-angle jump. The performance of proposed method along with the mentioned methods above will be compared under symmetrical and asymmetrical faults on a sample network. This network is a large scale real regional network including transmission and sub-transmission levels, which is modeled by using PSCAD/EMTDC. The output data are processed via MATLAB codes. The results determine the accuracy and validity of each method and show good performance of the proposed method and its unique applicability in cases where only currents are recorded. This study will help utilities in operation and network planning.

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1. Introduction

Voltage sag is a decrease to between 0.1 and 0.9 pu in rms voltage at the power frequency for durations of 0.5 cycle to 1 min [1]. This phenomenon is one of the most important PQ disturbances and due to its impact on sensitive industrial loads such as Adjustable Speed Drives (ASDs), Personal Computers (PCs) [2] and costs incurred by damage and maintenance, has captured the attention of utilities and researchers [3]. Voltage sag also can cause substantial loss of product of a typical industrial installation [4]. Starting of large induction motors, transformer saturation, capacitor switching and overloads may cause voltage sag, but faults (short circuits) are the main factors causing them. Occurred voltage sags are not confined to the fault location and propagate through the system affecting connected loads far away from the fault location [5]. Researches on voltage sags are focus on classification [6], detection [7], assessment [8] and representation of profiles [9]. But sag location is another important topic and will be the first step towards mitigation of PQ problems and it also plays a fundamental role in decision about responsibilities between the companies and customers and to determine financial penalties, in deregulated networks.

For voltage sag source location some studies have been done and several methods have been suggested. The first comprehensive work was presented in [10] in which by investigating the concepts of “disturbance power and energy”, fault location and capacitor switching causing disturbances were determined. In [11], Li et al.

determine the location of sag source by using the relation between measured voltage and current during sag. Later, the method was generalized by the same authors to “resistance sign” approach [12]. In this method locations of sag sources are found using the sign of obtained resistance from incremental voltage and current quantities. In [13], the polarity of variation for real current is used to classify the sag source as upstream or downstream. In [14], the sag source is found using seen impedance by distance relay and its angle. In [15], a method based on the information of the voltage magnitude and phase-angle jump was proposed to classify sag source as internal or external to the end-user grid at the connection point of sensitive customers. In [16], the method based on voltage information only was introduced. In [17], as an alternative method of [10], the variation of reactive power is investigated and in [18] the concept of phase change in sequence current (PCSC) was applied to locate faults in radial distribution systems. In [19] some of voltage sag source location methods have been surveyed and evaluated and their features and rules have been qualitatively and statistically analyzed.

This paper explains the methods principles for relative location of voltage sag sources first. Methods by using both voltage and current and voltage only and current only information have been included. The performance of these methods are compared in a large scale real power network under symmetrical and asymmetrical faults, in single source and two source systems and properties and defects of methods are provided all in detail. Moreover, in this paper a new method has been introduced based on the variation of the magnitude of positive-sequence current phasor and the sign of its phase-angle jump before and during the sag. This method is the only method capable of sag source location when voltage measurement

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is not available, because it uses the current information only (CBM¹). The proposed method is highly accurate and is more economic than methods with same accuracy.

2. Review on methods of voltage sag source location

The location of the sag source as observed from a monitored bus is defined as downstream (DS) region in the direction of the steady state active power flow or upstream (US) region in the opposite direction of it [20], as shown in Fig. 1.

2.1. Disturbance power and energy (DPE) method

The first comprehensive work for finding the source of the disturbances is based on the disturbance energy (DE) and the disturbance power (DP) [10]. The DP is the difference between the power delivered during the disturbance and during the normal condition. The DE results from the time integral of the DP. If the final value of the DE is greater than or equal to 80% of the peak value energy test is conclusive and the relative location of the disturbance source is determined, as shown in Fig. 2. Else, the energy test is inconclusive and the location of the disturbance source is determined by examining the polarity of initial peak of the DP. Its positive value indicates the DS source and negative value is indicative of the US source. Lack of mathematical analysis and the need for a threshold (80%) can be considered as disadvantages of this method.

2.2. Slope of system trajectory (SST) method

The method [11] is based on the relationship between the product of voltage magnitude and power factor against current magnitude shown in Fig. 1. For sample, for the DS fault (F_2), a line obtains with slope $-R_1$ as follows:

$$|V\cos\theta| = -R_1 I + E_1 \cos\theta_1 \quad (1)$$

Where E_1 is the voltage at the source 1, R_1 is the real part of Z_1 , V and I are the rms voltage and current, θ is the angle between voltage and current at the monitored location and θ_1 is the angle between I and E_1 . On the other hand, for the US fault (F_1), a similar linear equation is obtained with slope $+R_1$:

$$|V\cos\theta| = +R_1 I - E_1 \cos\theta_1 \quad (2)$$

Applying least-square method on points of $(I, |V\cos\theta|)$, a straight line is obtained. A negative slope indicates a DS fault and a positive slope means an US fault. Also an inversion of the direction of power flow during sag indicates an US fault. This method is evaluated using phase values. The need for points-fitting technique and lack of time scale are considered as disadvantages of this method.

2.3. Resistance sign (RS) method

The method [12] uses the sign of obtained resistance (R_e) from the following equations:

$$\begin{bmatrix} R_e \\ X_e \\ E_{Re} \end{bmatrix} = \begin{bmatrix} I_{Re}(1) & I_{Im}(1) & 1 \\ \vdots & \vdots & \vdots \\ I_{Re}(n) & I_{Im}(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_{Re}(1) \\ \vdots \\ V_{Re}(n) \end{bmatrix} \quad (3)$$

$$\begin{bmatrix} R_e \\ X_e \\ E_Y \end{bmatrix} = \begin{bmatrix} I_{Im}(1) & I_{Re}(1) & 1 \\ \vdots & \vdots & \vdots \\ I_{Im}(n) & I_{Re}(n) & 1 \end{bmatrix}^+ \begin{bmatrix} V_{Im}(1) \\ \vdots \\ V_{Im}(n) \end{bmatrix} \quad (4)$$

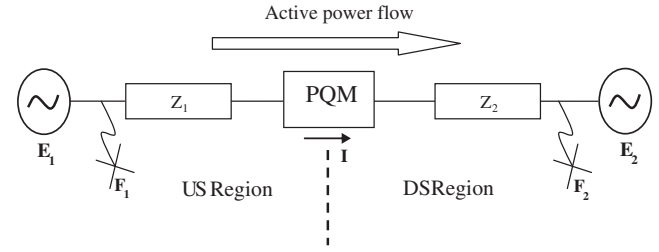


Fig. 1. Basic circuit to describe the localization of the sag source as upstream or downstream.

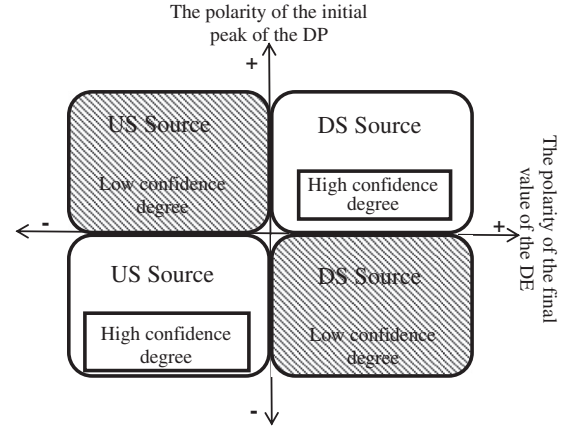


Fig. 2. Disturbance source location using energy test.

where the symbol of “+” indicates pseudo-inverse. If two estimated resistances using (3) and (4) have the same sign the test is conclusive. A positive sign indicates a US fault whereas a negative sign means a DS fault. If the power flow is reversed during the sag, the data of positive-sequence voltages and currents (n) used in (3) and (4) must only include values before the reversion of the power flow. Otherwise if the power flow does not change the value of n should be selected as the sag duration. However, the results of the method in [20] show that this method is very sensitive to the voltages and currents values chosen and in many of the simulated cases it fails. Therefore, it will not be investigated in our paper.

2.4. Real current component (RCC) method

In this method [13], locations of sag sources are found using the polarity of the “variation” of the real part of the current. For this, coordinates of $(I\cos\theta)$ before and during the sag should be recorded and then plotted against time and its polarity at the beginning of the fault is determined. A positive polarity indicates a DS fault whereas a negative polarity means a US fault. In this method the value of the fault impedance is taken into consideration. Also like to [12], this method is evaluated using phase values.

2.5. Distance relay (DR) algorithm

This algorithm is based on the principle that the magnitude and angle of impedance before and during the sag event clearly indicate the relative location of the sag source with respect to the PQ monitor [14]. For this purpose, the DR algorithm uses DR information for sag source location. For the DS fault (F_2) in Fig. 1, the seen impedance at the PQ monitor will be:

$$Z_{PQM} = \frac{V}{I} = Z_{2-Pos.Seq} + \Delta Z \quad (5)$$

¹ Current Based Method.

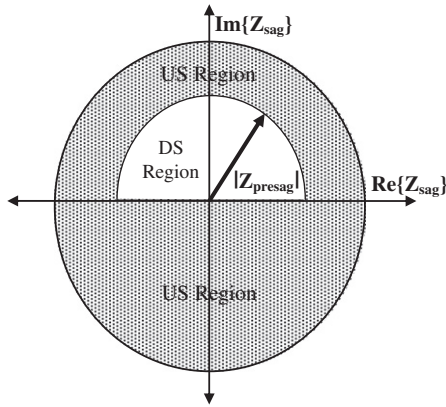


Fig. 3. The impedance-plane representing the associated DR rule logic.

where V and I are the measured voltage and current phasors, ΔZ is a function of fault resistance, load angle, etc. For US faults (F_1), the resulting impedance will change in both magnitude and angle. The DR rule for sag source location becomes: if $|Z_{sag}| < |Z_{presag}|$ and $\angle Z_{sag} > 0$ then sag source is in DS of the PQ monitor else, the sag source is in US. The impedance-plane representing the associated above rule logic is in Fig. 3.

2.6. The method based on the sag magnitude and its phase-angle jump (VBM²)

The method [15] states that for a typical industrial installation the sags generated in the utility grid and the sags generated inside the industrial grid will follow different phase-angle jump vs. magnitude relations, and that the difference is larger for deeper sags. Therefore, in this case, it would be a good index for finding voltage sag source locations. However, this method is not suitable for classifying sags at the interconnection point of two utilities at transmission levels because it is not expected to observe such a phase-angle jump vs. magnitude characteristic when there are transmission networks at both sides of the monitoring point; therefore, it will not be investigated in our paper.

2.7. Voltage sag magnitude (VSM) method

The idea of this method [16] is to compare the voltage sag magnitudes in per unit at primary side (V_1) and at secondary side (V_2) of the transformer at transmission and sub-transmission levels (see Fig. 4). For the DS fault (F_2) in the system shown in Fig. 4, the voltage drop at each side of the transformer is given by:

$$\Delta V_1 = Z_1 I \quad (6)$$

$$\Delta V_2 = (Z_1 + Z_{TRAFO}) I \quad (7)$$

where Z_{TRAFO} is the transformer impedance and I is the total of fault and load currents. The voltage drop is higher on the side where the fault is located. Therefore, if $\Delta V_2 > \Delta V_1$ (i.e. $V_1 > V_2$) the fault is DS, otherwise if ($V_1 \leq V_2$) the fault is US. If the DS system does not include generation units such as DGs, for US faults, $V_1 = V_2$.

2.8. Reactive power (RP) method

In [17], the variation of reactive power is investigated. According to the author's viewpoint regardless of whether the monitoring point is operating with positive or negative power, the location criteria can be expressed thus: if there is a positive variation of

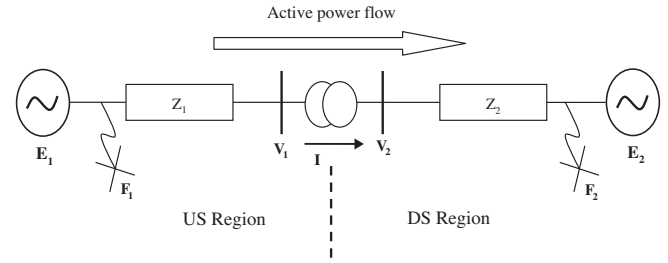


Fig. 4. Basic circuit to describe the voltage sag source location using sag magnitude.

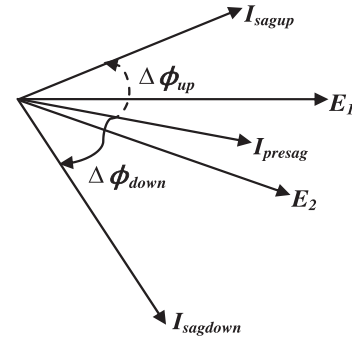


Fig. 5. Phasor diagram of the network shown in Fig. 1.

reactive power during sag ($\Delta Q > 0$) and $\tan \theta = \frac{Q}{P} > 0$ then a fault is considered DS. Else if ($\Delta Q < 0$) or $\tan \theta < 0$ the fault is in US.

2.9. Phase change in sequence current (PCSC) method

This method [18] finds the location of sag source using the angle difference of the before and during the sag positive-sequence component of current. Fig. 5 shows the phasor diagram of the power network shown in Fig. 1, where I_{sagup} and $I_{sagdown}$ are currents measured at the PQ monitor for US and DS faults. I_{presag} corresponds to the current before the sag event, and $\Delta \phi_{up} = \angle I_{sagup} - \angle I_{presag}$ and $\Delta \phi_{down} = \angle I_{sagdown} - \angle I_{presag}$. E_1 and E_2 are the voltages at sources 1 and 2, respectively. In Fig. 5, it can be seen that the angle difference of the I_{sagup} and I_{presag} phasor for US faults is positive ($\Delta \phi > 0$), and for DS it is negative ($\Delta \phi < 0$). The PCSC rule for sag source location becomes: If $\Delta \phi > 0$ then sag source is in US else, if $\Delta \phi < 0$ sag source is in DS. Where $\Delta \phi$, is the angle difference of during and before sag positive-sequence component of current (positive-sequence current phase-angle jump) and is set within $-\pi$ to π .

3. Proposed method, Current Based Method (CBM)

This method is based on variation of the magnitude and the sign of the phase-angle jump of the passing positive-sequence current of the DOC relay in the before and during the sag condition. For this purpose, the proposed method uses DOC relay information or employing its algorithm for sag source location. For the DS fault (F_2) in the system shown in Fig. 1, according to DOC relay, below calculations can be done for tripping the circuit at the PQ monitor:

$$|I| > I_{pickup} \quad \text{and} \quad -\pi < \angle I < 0 \quad (8)$$

where I_{pickup} is the relay setting current and $\angle I$ is angle between reference phasor and fault current. For US faults (F_1), the current direction will be reversed and the passing current will change in both magnitude and angle. Therefore, the magnitude and angle of the current computed from current phasor has a distinctive feature in identifying the faults in front of or behind the relay/PQ monitor. Considering the measured voltage at the relay/PQ monitor as a

² Voltage Based Method.

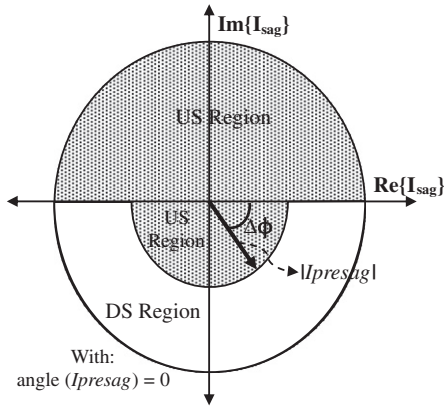


Fig. 6. Current-plane representing the associated proposed rule logic.

polarizing quantity and reference phasor, the rule for sag source location becomes as follows:

If $|I_{sag}| > |I_{presag}|$ and $\angle I_{sag} < 0$ then the sag source is in DS of the PQ monitor
 else if $|I_{sag}| \leq |I_{presag}|$ or $\angle I_{sag} > 0$ the sag source is in US. (9)

In above rule, $\angle I_{sag}$ is the current phasor angle during sag set within $-\pi$ to π . I_{sag} and I_{presag} are positive-sequence current phasor during sag and presag in monitoring point respectively. Now instead of using the $\angle I_{sag}$, positive-sequence current phase-angle jump (PCSC ($\Delta\phi$)) [18] is used and our proposed method is formed based on only current information. The rule for voltage sag source location becomes:

CBM Rule :If $|I_{sag}| > |I_{presag}|$ and $\Delta\phi < 0$ then the sag source is in DS of the PQ monitor
 else if $|I_{sag}| \leq |I_{presag}|$ or $\Delta\phi > 0$ the sag source is in US. (10)

where $\Delta\phi = \angle I_{sag} - \angle I_{presag}$ and is set within $-\pi$ to π . The current-plane representing the associated above rule logic is in Fig. 6, (with $\angle(I_{presag}) = 0$). Area below the real axis and outside of the inner semicircle shows DS regions and the rest identifies US regions.

It can be seen that CBM method is totally based on the current. By comparing CBM method rule with DR method [14], it is found that CBM method is using during sag and presag current phasors while the DR method is using during sag and presag phasors of both the current and voltage. The difference in the criterion of DOC relay and the proposed method is that DOC relays do not use I_{presag} but the proposed method uses prefault current phasors which have applications in digital relaying for fault detection, fault classification, and other applications [21].

The proposed method is tested to locate voltage sag sources at transmission and sub-transmission levels (PQ monitors in this method use the estimates of the DOC relays (or its algorithms)). At these levels the influence of loads is not so evident. And the effect of constant power loads may not affect the performance of the method because other loads such as induction machines will supply the extra current demanded by the constant power loads during short sags [22].

4. Simulations

4.1. Test power system

Some parts of the Brazilian power network (Mato Grosso³) were selected for testing methods (see Fig. 7). This real network has been

modeled by PSCAD/EMTDC V 4.2.0 and the output data are processed via MATLAB 7.0 codes. The network includes transmission and sub-transmission levels, as well as meshed and radial system configuration at frequency of 60 Hz. Also the system is highly interconnected and there is a considerable amount of Distributed Generation (DG) units. The features of this system are provided in Table 1.

4.2. Monitors, active power flow, upstream and downstream

Radial parts of the power network have been used to test methods and there are both single source and two source systems as shown in Fig. 7a and b, respectively. It can be seen from these figures that M_1 and M_2 monitors are between the radial sub-transmission network and transmission network. The M_1 monitor has been installed at the border of a single source radial network and the M_2 monitor has been installed in border of a two source radial network due to the presence of a DG in DS (dash ellipse in Fig. 7b). The 230 kV transmissions and the 138 kV sub-transmission networks are managed by two different utilities. Because of this, sag source location needs to be done considering the borders between two utilities. In this test network the active power flows from the 230 kV system to the 138 kV system in monitored buses shown by blue arrows in Fig. 7a and b. Therefore, US is equivalent to outside and DS is equivalent to inside when the 138 kV system is taken as the reference.

4.3. Characteristic of the applied faults

Because system faults (short circuits) are the main reason resulting in the voltage sags, the voltage sag source location in this paper is studied from the point of view of the line faults. Ten fault locations ($F_1 \dots F_{10}$) shown in Fig. 7, and in each location four different types of faults were simulated. LLLG, LG in phase-a, LLG and LL in b and c are the types of faults. The resistance of the faults was 1 Ω and with duration of 0.1 s.

4.4. Results

In the first part of this section, simulation results of all the previous methods except for the PCSC are presented. Results from the PCSC and the proposed (CBM) methods will be mentioned later. The results for both single and two sources radial systems corresponding to M_1 and M_2 in Brazilian power network are presented simultaneously. In each part there are two cases; the results for symmetrical faults (LLLG) and asymmetrical faults (LL, LG and LLG) respectively.

4.4.1. Results of previous methods

4.4.1.1. Symmetrical faults. Firstly, results of the faults located at F_1 (in bus 205) and F_8 (in bus 2018) will be presented and discussed. The location of F_1 fault is at US for both M_1 and M_2 monitors. F_8 is located at US for M_1 and at DS for M_2 . Fig. 8 shows the results of DPE method for a fault at F_8 location. The final DE at M_1 monitor is negative, therefore the fault location is successfully found at US. The final DE at M_2 monitor is also negative which shows a US fault, but the initial peak of DP is positive that reduces the confidence degree of the method. The real location of the fault is DS so this monitor had not worked correctly anyway.

In Fig. 9, the SST method has been applied and the results for F_8 location were presented. At M_1 , the slopes for all three phases are positive which show a US fault. At M_2 , the slope is negative in all three phases which indicates a DS fault. The results for RCC method for the fault at F_1 have been presented in Fig. 10. All three phases at both monitors have negative polarities which show a US fault.

Results of VSM method for F_1 and F_8 locations were shown in Table 2. For example, for a fault at F_8 at M_2 , the sag magnitude at

³ Mato Grosso is one of the biggest states in the western part of Brazil.

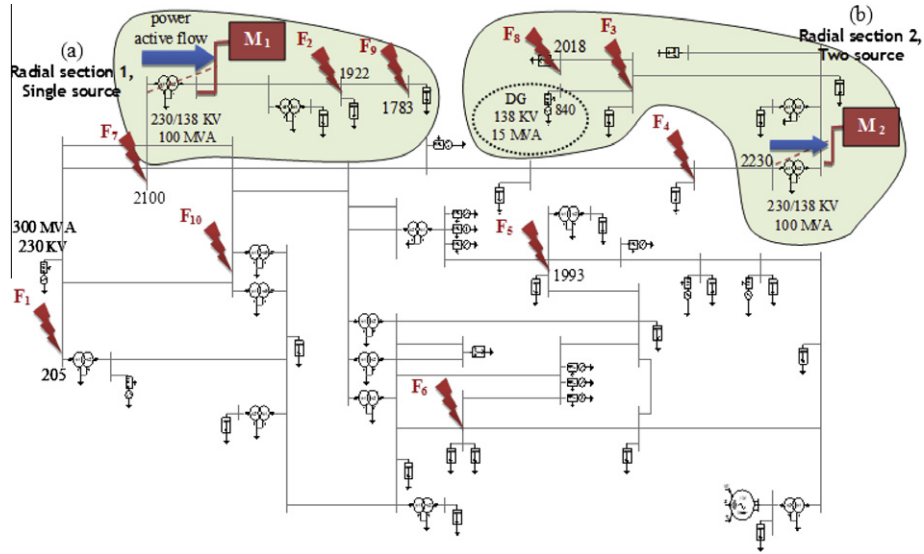


Fig. 7. Brazilian power network (Mato Grosso) one-line diagram, fault locations ($F_1 \dots F_{10}$), monitors (M_1, M_2) and direction of active power flow (blue arrows) are indicated. Radial Section 1 (i.e. single source system) at (a) and radial Section 2 (i.e. two source system) at (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Features of Brazilian power network (Mato Grosso).

Buses of radial sections	Generation capacity/ present demand	Export of power from:	Transformers winding	Transformer capacity (MVA)	No. of substations	Lines length (230,138 KV) (km)	No. of lines
From 2100 to 1783 and from 2230 to 840	1643 MVA/690 MW with high load factor	Bus 205		2076	93	6619 with line max. length of 429 km	67

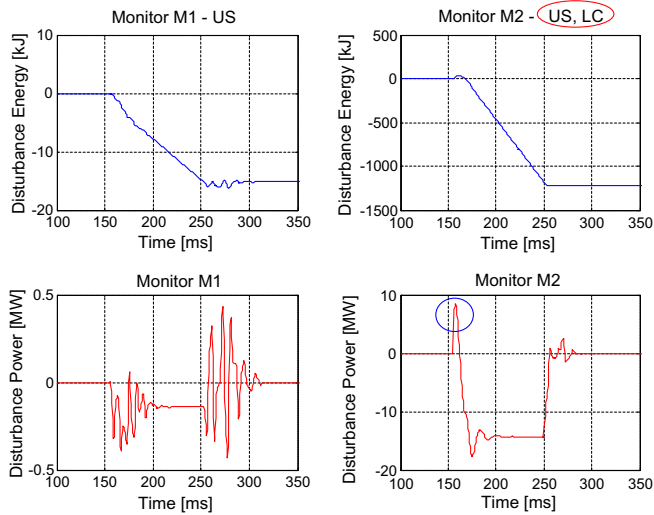


Fig. 8. Results of the DPE method, LLLG fault at F_8 .

230 kV bus is 0.69 per unit whereas it is 0.6 per unit at 138 kV bus (i.e. $V_1 > V_2$) which is an indication of a DS fault.

Table 3 shows the results of DR method for faults at F_1 and F_8 . DR method has worked correctly for the both fault locations at each monitor. For example, for the fault at F_1 , due to the impedance remaining constant of 1324 Ω and the angle of -40° , M_1 shows a US fault. The results of the RP method for the fault at F_1 are shown in Fig. 11. At M_1 , during the disturbance $\Delta Q > 0$ and $Q < 0$ but for $\tan \theta = \frac{Q}{P} < 0$, the fault is recognized at US. For M_2 , during the disturbance $\Delta Q < 0$ which also shows a US fault.

4.4.1.2. Asymmetrical faults. In this section for instance the results of the SST method for a LLG fault in b and c phases at F_1 are presented. Fig. 12 shows it. Considering the faulted phases as an index in this case, both phases have positive slopes at M_1 which is an indication of a US fault. At M_2 each phase has a different slope and consequently the method is indecisive for this case.

4.4.2. Results of the PCSC and the proposed (CBM) methods

In this section results of the PCSC and the CBM methods are discussed. To ensure the reliability of the CBM method, a certain minimum magnitude variation of current is considered. If the difference between $|I_{sag}|$ and $|I_{presag}|$ is less than 5 A, then $|I_{sag}| = |I_{presag}|$ (no increase no decrease) is implied (at both US and DS faults).

4.4.2.1. Symmetrical faults. Firstly, the results of faults located at F_1 , F_2 , F_5 and F_8 will be presented and discussed. The F_2 location is DS with respect to M_1 and US to M_2 . F_5 is located US for both M_1 and M_2 . All the results are presented in Figs. 13–16. In Fig. 13, results of PCSC and CBM methods for symmetrical fault at F_1 location are demonstrated. In this figure, at M_1 , the positive-sequence current magnitude during sag has decreased and its phase-angle jump is positive ($\Delta\phi > 0$), thus, CBM and PCSC methods show an US fault. At M_2 the current magnitude has remained constant during sag and the phase-angle jump is positive ($\Delta\phi > 0$), and both methods show US fault.

Fig. 14 shows a symmetrical fault at F_2 location. In this figure at M_1 , the current magnitude has raised and its phase-angle jump is negative ($\Delta\phi < 0$) which is an indication of a DS fault for PCSC and CBM methods. At M_2 , the current magnitude has not changed and its phase-angle jump is positive ($\Delta\phi > 0$) which shows the US

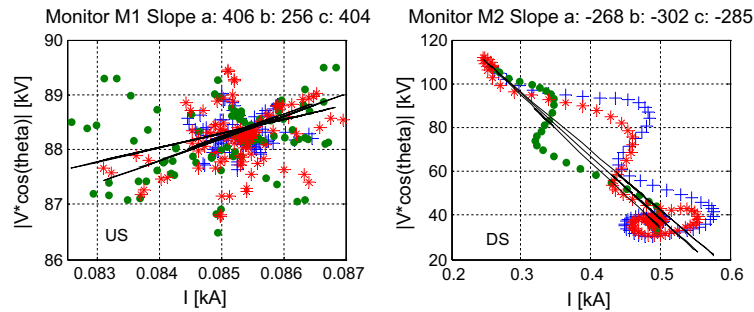
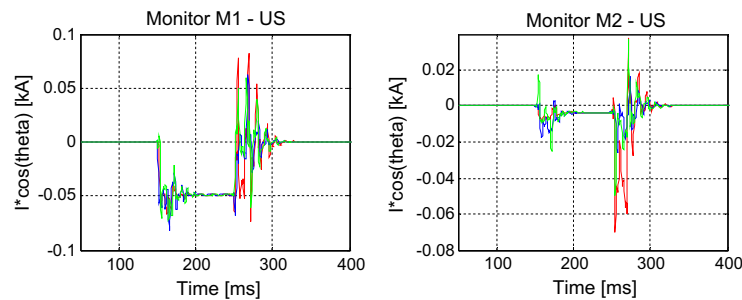
Fig. 9. Results of the SST method, LLLG fault at F_8 .Fig. 10. Results of the RCC method, LLLG fault at F_1 .

Table 2
Results of the VSM method, LLLG faults at F_1 and F_8 .

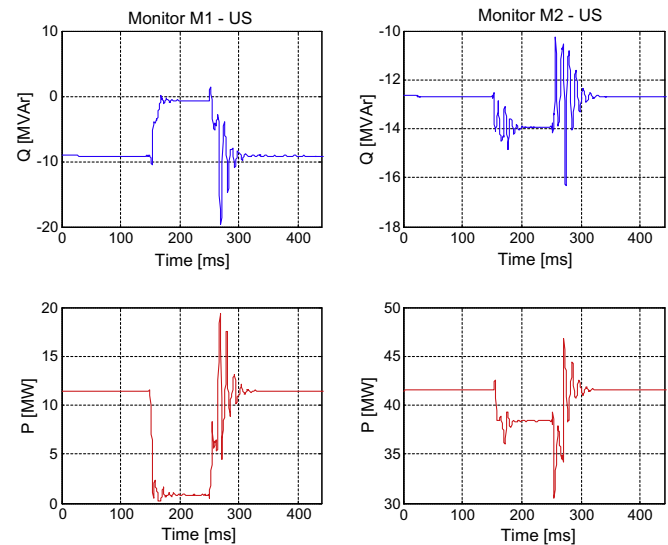
Faults	F_1 (LLLG)		F_8 (LLLG)	
	Monitors		Monitors	
	M_1	M_2	M_1	M_2
Buses (kV)	230/138	230/138	230/138	230/138
V sag	0.27/0.27	0.93/0.94	0.99/0.99	0.69/0.6
VSM Result	US	US	US	DS

US: Upstream, DS: downstream.

Table 3
Results of the DR method, LLLG faults at F_1 and F_8 .

Faults	F_1 (LLLG)		F_8 (LLLG)	
	Monitors		Monitors	
	M_1	M_2	M_1	M_2
$ Z_{presag} $ (Ω)	1324	436	1324	436
$ Z_{sag} $ (Ω)	1324	408	1324	139
$\angle Z_{sag}$ ($^\circ$)	-40	-20	-38	56
DR result	US	US	US	DS

US: Upstream, DS: downstream.

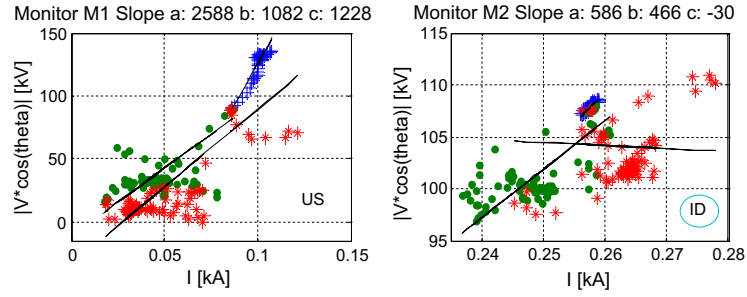
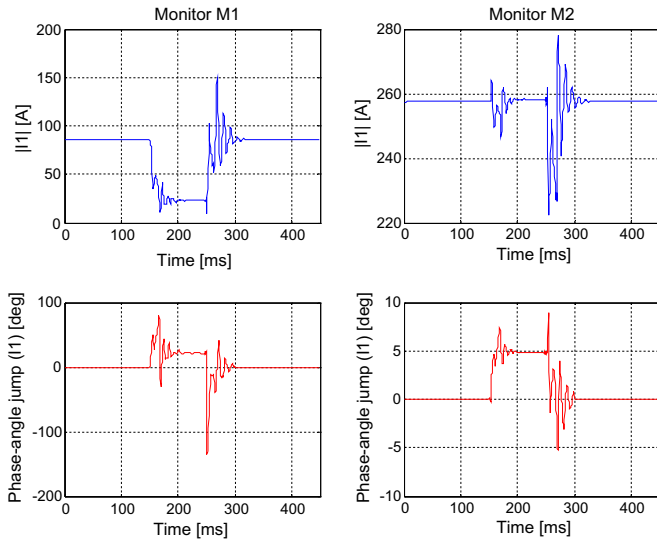
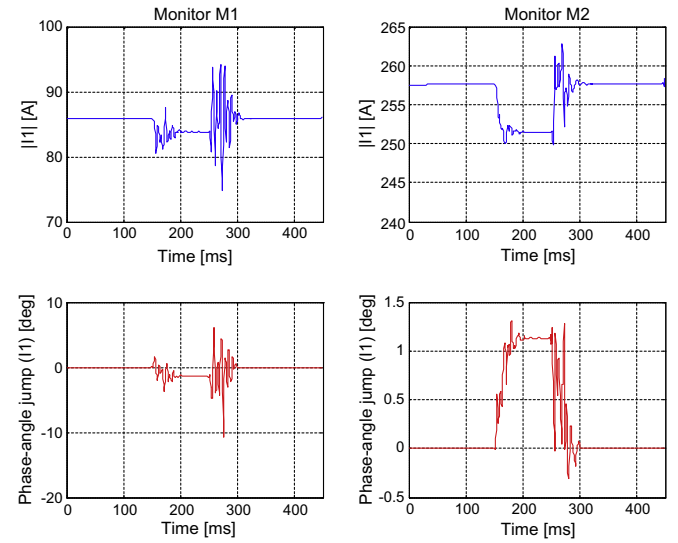
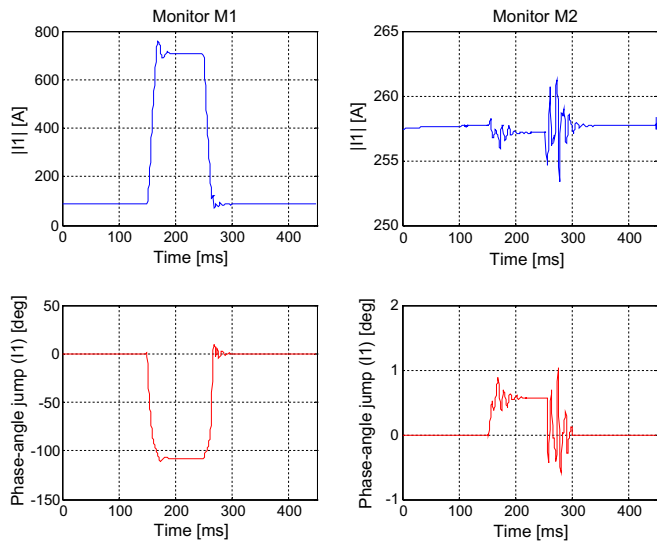
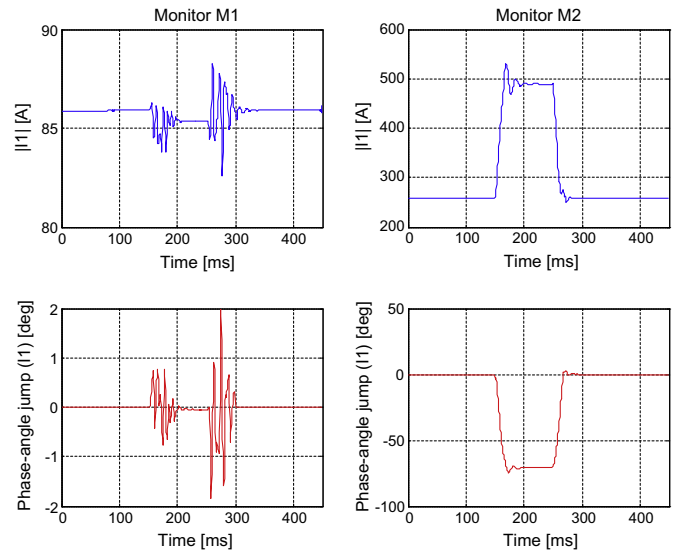
Fig. 11. Results of the RP method, LLLG fault at F_1 .

fault for both methods. In Fig. 15, results for fault at F_5 are shown. At M_1 , the magnitude of the current has not been changed and the CBM method shows a US fault. In this monitor the PCSC method shows a DS fault because ($\Delta\phi < 0$), which is incorrect. At M_2 , the phase-angle jump is positive ($\Delta\phi > 0$), consequently both methods show US fault.

Fig. 16 shows a symmetrical fault at F_8 . At M_1 the current magnitude is almost fixed and regardless of the phase-angle jump, the CBM method shows an US fault. The PCSC method is indecisive at M_1 because the ($\Delta\phi \approx 0$) is obtained. At M_2 the current magnitude has increased and the phase-angle jump is negative ($\Delta\phi < 0$) which shows a DS fault for both the CBM and the PCSC methods.

4.4.2.2. Asymmetrical faults. Results of asymmetrical faults at F_1 , F_2 , F_5 and F_8 are organized in Tables 4–7 respectively. The first column shows real location of the faults for both monitors (according to Fig. 7), and the rules for the two methods. The second and the third columns are the results for LG fault, fourth and fifth columns show the result for LLG fault and sixth and seventh columns are also presenting the result for LL fault.

In Tables 4, 6 and 7, the CBM method has identified the fault locations correctly in all cases, but the PCSC method has not worked correctly in identifying almost all the US faults at M_1 which is related to a single source radial transmission network. For instance, for a LG fault at F_1 location at M_1 , the magnitude of the positive-sequence current has changed from 85.8 A to 84 A (the difference is less than 5 A, therefore $|I_{sag}| = |I_{presag}|$). These changes

Fig. 12. Results of the SST method, LLG fault at F_1 .Fig. 13. Positive-sequence current (I_1) and phase-angle jump plot, LLG fault at F_1 .Fig. 15. Positive-sequence current (I_1) and phase-angle jump plot, LLG fault at F_5 .Fig. 14. Positive-sequence current (I_1) and phase-angle jump plot, LLG fault at F_2 .Fig. 16. Positive-sequence current (I_1) and phase-angle jump plot, LLG fault at F_8 .

show a US fault for the CBM method but the phase-angle jump of -0.75° indicates a DS fault incorrectly for the PCSC method. For a LLG fault at F_5 at M_2 , the magnitude of the current has not changed (the difference is less than 5 A, therefore $|I_{sag}| = |I_{presag}|$), and its phase-angle jump is $+0.6^\circ$, as the result the CBM method and also the PCSC method show a similar US fault.

In Table 5, the CBM method has successfully found the location of the voltage sag source similar to the PCSC method. For example, for a LL fault at F_2 at M_1 , the current magnitude has risen from 86.1 A to 343 A, and its phase-angle jump is -101° which indicates a DS fault for the CBM and the PCSC methods.

Table 4
Results of the PCSC and the CBM methods, (LG, LLG and LL) faults at F₁.

Faults	F ₁ (LG)		F ₁ (LLG)		F ₁ (LL)	
	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
Monitors [US/US]	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
$ I_{presag} $ (A)	85.8	257.4	85.8	257.4	85.8	257.4
$ I_{sag} $ (A)	84	257.2	53.4	257.5	54	257.8
$\Delta\phi$ (°)	−0.75	0.15	8.3	2.44	9.3	2.42
PCSC result	DS ^a	US	US	US	US	US
CBM result	US	US	US	US	US	US

US: Upstream, DS: downstream.

^a Wrong operation.

Table 5
Results of the PCSC and the CBM methods, (LG, LLG and LL) faults at F₂.

Faults	F ₂ (LG)		F ₂ (LLG)		F ₂ (LL)	
	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
Monitors [DS/US]	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
$ I_{presag} $ (A)	86.2	257.4	86.1	257.4	86.1	257.4
$ I_{sag} $ (A)	130	257.5	384	257.3	343	257.4
$\Delta\phi$ (°)	−66.3	0.9	−101	0.3	−100	0.29
PCSC result	DS	US	DS	US	DS	US
CBM result	DS	US	DS	US	DS	US

US: Upstream, DS: downstream.

Table 6
Results of the PCSC and the CBM methods, (LG, LLG and LL) faults at F₃.

Faults	F ₃ (LG)		F ₃ (LLG)		F ₃ (LL)	
	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
Monitors [US/US]	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
$ I_{presag} $ (A)	85.8	257.4	85.8	257.4	85.8	257.4
$ I_{sag} $ (A)	85.6	257	84.8	254.2	84.8	254.6
$\Delta\phi$ (°)	−0.6	0.35	−0.77	0.6	−0.8	0.6
PCSC result	DS ^a	US	DS ^a	US	DS ^a	US
CBM result	US	US	US	US	US	US

US: Upstream, DS: downstream.

^a Wrong operation.

Table 7
Results of the PCSC and the CBM methods, (LG, LLG and LL) faults at F₈.

Faults	F ₈ (LG)		F ₈ (LLG)		F ₈ (LL)	
	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
Monitors [US/DS]	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
$ I_{presag} $ (A)	85.8	257.7	85.8	257.7	85.8	257.7
$ I_{sag} $ (A)	85.8	265.2	85.6	332.7	85.6	311
$\Delta\phi$ (°)	−0.2	−18.4	−0.3	−51	−0.3	−48
PCSC result	DS ^a	DS	DS ^a	DS	DS ^a	DS
CBM result	US	DS	US	DS	US	DS

US: Upstream, DS: downstream.

^a Wrong operation.

As it can be seen from Tables 4–7, the obtained phase-angle jump ($\Delta\phi$) is small for the remote faults of monitors such as for F₁ and F₂ at M₂ and F₈ at M₁. Whilst for closer faults such as the

Table 8
Results of the CBM method for LG faults at locations of F₂ and F₇ with neutral grounding resistances of 0, 0.5, 1 and 5 Ω . Fault resistance is 1 Ω .

Faults – location	F ₂ (LG) – [DS for M ₁ /US for M ₂]								F ₇ (LG) – [US for M ₁ /US for M ₂]							
	0 (Ω)		0.5 (Ω)		1 (Ω)		5 (Ω)		0 (Ω)		0.5 (Ω)		1 (Ω)		5 (Ω)	
Monitors	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂	M ₁	M ₂
$ I_{presag} $ (A)	86.2	257.4	86	257.4	86	257.4	85.8	257.4	85.8	257.4	85.8	257.4	85.8	257.4	85.8	257.4
$ I_{sag} $ (A)	130	257.5	129.35	257.4	129	257.6	127.35	257.5	80.6	256.4	77.2	256	74.5	256.3	54.3	257.5
$\Delta\phi$ (°)	−66.3	0.9	−71	1	−75.5	1	−74.5	1.05	−5.7	3.3	−5.2	2.12	−3.7	1.64	−3.5	1.05
CBM result	DS	US	DS	US	DS	US	DS	US	US	US	US	US	US	US	US	US

US: Upstream, DS: downstream.

Table 9
LLG faults at different locations.

Fault/location	DPE	SST	RCC	DR	VSM	RP	PCSC	CBM
<i>Monitor M₁</i>								
F ₁ /US	US	US	US	US	US	US	US	US
F ₂ /DS	DS	DS	DS	DS	DS	DS	DS	DS
F ₃ /US	US	US	US	US	US	US	DS ^a	US
F ₄ /US	US	US	US	US	US	US	DS ^a	US
F ₅ /US	US	US	US	US	US	US	DS ^a	US
F ₆ /US	US	US	US	US	US	US	DS ^a	US
F ₇ /US	US	US	US	US	US	US	DS ^a	US
F ₈ /US	US	US	US	US	US	US	ID ^b	US
F ₉ /DS	DS	DS	DS	DS	DS	DS	DS	DS
F ₁₀ /US	US	US	US	US	US	US	DS ^a	US
<i>Monitor M₂</i>								
Fault/location	DPE	SST	RCC	DR	VSM	RP	PCSC	CBM
F ₁ /US	US	ID ^b	US	US	US	US	US	US
F ₂ /US	US	US	US	US	US	US	US	US
F ₃ /DS	US-LC ^a	DS	DS	DS	DS	DS	DS	DS
F ₄ /US	US	DS ^a	US	US	US	US	US	US
F ₅ /US	US	US	US	US	US	US	US	US
F ₆ /US	US	US	US	US	US	US	US	US
F ₇ /US	US	US	US	US	US	US	US	US
F ₈ /DS	US-LC ^a	DS	DS	DS	DS	DS	DS	DS
F ₉ /US	US-LC ^a	US	US	US	US	US	US	US
F ₁₀ /US	US	US	US	US	US	US	US	US

US: Upstream, DS: downstream, ID: indecisive, LC: low confidence.

^a wrong operation.

^b ID test.

F₁ and F₂ at M₁ and F₈ at M₂, the results of ($\Delta\phi$) is more reliable and give exact information for its current signal. (F₅ is located almost between monitors).

4.4.2.2.1. Influence of neutral grounding resistance on the proposed (CBM) method. The results of proposed method for LG faults considering neutral grounding resistances of 0, 0.5, 1 and 5 Ω are summarized in Table 8. As expected, the grounding resistance as well as fault resistance affect and limit the sequence currents of faults during LG faults. These changes are only for the results of the very close faults to the monitors (M₁ and M₂ in Fig. 7). It can be noted from Table 8 that for the LG faults at F₇ location which is very close to M₁, increasing grounding resistance decreases the magnitude of positive-sequence current and changes the ($\Delta\phi$), but it does not affect the decisions on our proposed method (CBM) and the fault is correctly located US. For the faults happened exactly at the monitored buses even if the proposed method is failed, it is not important because according to the concepts of US/DS the duty of this monitor is not to recognize this kind of faults. The other monitors in the system are responsible for recognizing these fault locations. As the result, the faults on monitored buses also do not affect the accuracy of proposed method.

The grounding resistance does not change the results for remote faults in monitors. For instance, for the LG faults at F₂ in both monitors and the F₇ fault at M₁, increasing value of grounding resistance does not change the magnitude of positive-sequence current and the phase changes ($\Delta\phi$), therefore in these cases, the proposed method works well.

The proposed method is similar to the previous methods of being able in working in a radial system. In the studied power system, there are two 138 kV sub-transmission radial sections (Fig. 7 and Table 1). Transformer 230/138 kV connects the 230 kV transmission network to each of radial sections. The transmissions and the sub-transmission networks are managed by two different utilities. Because of this, sag source location needs to be done considering the borders between the two utilities. Therefore the best locations of monitors are in the same borders between the two utilities (M_1 and M_2 in Fig. 7).

As a result, the proposed method can be implemented in transmission and sub-transmission levels successfully without being influenced by neutral grounding resistance.

4.4.3. Results of all the methods for 10 fault locations

To test the performance and sensitivity of all methods for symmetrical and asymmetrical faults, 10 more faults were simulated.

Table 9 shows the results of these simulations under symmetrical faults for 10 fault locations ($F_1 \dots F_{10}$). First column of the table indicates real location of the fault and other columns show the performance of different methods. It can be seen that at M_1 monitor, for most cases of the US faults, PCSC method has not worked properly.

Symmetrical faults are the most sever faults in any power systems [23], but as shown in Fig. 17, most methods were responded to these types of faults correctly. In symmetrical faults, the correct results for both monitors in different methods are as follows (out of 20 fault): PCSC method 13 correct responses (65%), DPE method 17 correct responses (90%) and RCC, DR, VSM, RP and CBM methods 20 correct responses (100%).

The results of asymmetrical faults for M_1 monitor in all 10 fault locations are reported distinctively in Fig. 18. The results for M_2 monitor have been reported in Fig. 19. In cases which the DPE method has found the fault location with low confidence degree, it has also been considered as wrong fault location identification. In RCC and SST methods in which the results are derived by analyzing each phases, at LLG and LL faults, the results for the faulted phases may be different which makes this test indecisive. PCSC method at M_1 has the worst performance which shows that this method is not appropriate for the fault relative location (US) in single source transmission networks. LG faults are the most frequent faults in power systems [23]. It can be seen from Figs. 18 and 19 that all methods have better performances under LG faults in comparison with LLG and LL faults which are not happened as frequently as the LG faults.

The results of all the methods for all the asymmetrical faults at M_1 and M_2 monitors have been shown quantitatively in Fig. 20. In Asymmetrical faults, the correct results of each method for both monitors are as follows (out of 60 fault): RCC method 34 correct responses (56.66%), PCSC method 38 correct responses (63.33%), DPE method 40 correct responses (66.66%), SST method 47 correct responses (78.33), RP method 59 correct responses (98.33%) and the methods of CBM, DR and VSM 60 correct responses (100%).

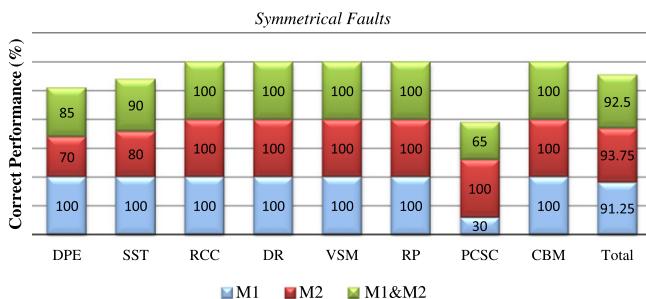


Fig. 17. Results of all the methods quantitatively for symmetrical faults.

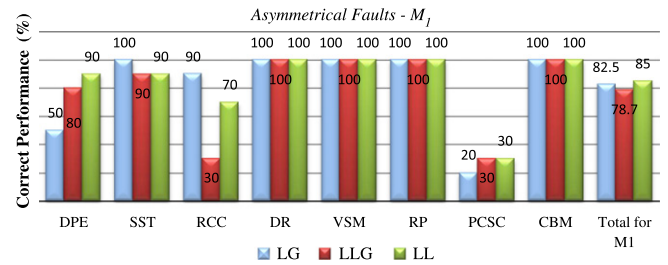


Fig. 18. Results of all the methods separately for all the asymmetrical faults at M_1 monitor.

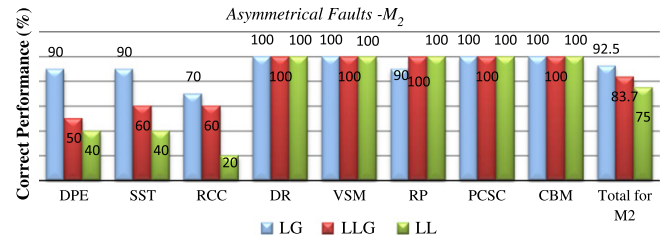


Fig. 19. Results of all the methods separately for all the asymmetrical faults at M_2 monitor.

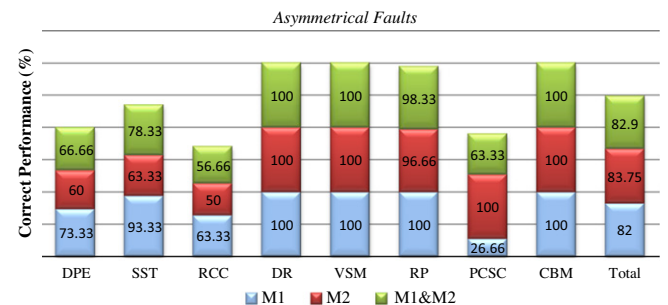


Fig. 20. Results of all the methods quantitatively for all the asymmetrical faults.

5. Discussion

In this section, the comprehensiveness of this paper will be expressed first and then all other methods and their corresponding simulation results will be compared in terms of the accuracy, hardware costs, ease and speed of performance, effect of faulted phases and the presence of DG.

5.1. Comprehensiveness of this paper

In [20], the performance of five methods under six fault locations ($F_1 \dots F_6$) have been studied in both symmetrical and asymmetrical (only the LG fault) in Brazilian power network. Whereas in this paper, the performance of eight methods under 10 fault locations ($F_1 \dots F_{10}$) are evaluated either symmetrically or asymmetrically (the LL, LLG and LG faults), one of these methods is also new, so this study can be considered to be comprehensive.

5.2. Accuracy and hardware cost

In terms of accuracy the methods are categorized as follows; for the symmetrical faults: (CBM, RCC, DR, VSM and RP), SST, DPE and PCSC. For the asymmetrical faults we have: (CBM, VSM and DR), RP, SST, DPE, PCSC and RCC. Overall, the total number of correct results for all previous methods along with the proposed method

(CBM) for symmetrical faults (in both monitors) are the 148 correct results out of 160 (92.5%) and for asymmetrical faults are the 398 correct results out of 480 (83%). The proposed method (CBM), VSM and DR methods have shown the best performances with 100% correct results for both symmetrical and asymmetrical. The VSM method is economical in term of hardware costs because it requires only one variable to measure, but in distribution levels in which the influence of loads is evident, this method may not work properly. The DR method uses both voltage and current variables which leads to higher costs. But if it is applied in distributions levels, it is still capable of finding the sag source location correctly. The CBM method requires only the current variable which is also quite economical. If this method is applied in distribution levels, it will work well because the phase-angle jump ($\Delta\phi$) [18] acts properly in the levels even if the magnitude of positive-sequence current does not act appropriately.

5.3. Ease of implementation and speed of run

Comparing with the other methods which measure the current and the voltage, the VSM method uses voltage measurements only and the PCSC and CBM use the current measurements. Therefore, these methods are better in terms of the ease of implementations. Also the speed of running for VSM, PCSC and CBM is higher compared with other methods.

5.4. Effect of faulted phase on the correctness of responses

Results for DR, RP, DPE, PCSC, VSM and CBM methods remain unchanged (with a change in the faulted phase), because the outputs are not based on per phase. As an instance, simulation of a LG fault at a particular point in a, b and c phases has revealed identical results for impedance and its angle in DR method. Similarly, for LL and LLG faults simulation in ab, bc and ca phases, the results are approximately identical. A LLL and LLLG faults has also similar results. But in RCC and PCSC methods, the simulation for faults in different phases produces different results.

5.5. Effects of DG

Due to existence a DG in bus 840, a two source network will be made. In this network, for M_2 monitor with US faults, there will be a high probability of reversing the current direction and consequently some results may show more inaccuracy. Results showed that the SST method may not work properly for US faults in two sources networks, but for DS faults the method will work properly since the current direction will remain unchanged during the fault.

Results from simulations also showed that in single source radial transmission networks where the DG is not installed in DS region of M_1 monitor, the PCSC method does not work properly for US faults. In using the proposed method (CBM), the performance of PCSC method has been enhanced greatly in transmission and sub-transmission levels. This has been done by adding positive-sequence current to PCSC method. The results also stated that the proposed method (CBM) is able to locate voltage sag source even with the presents of a grounding resistance.

6. Future works

- (1) For power system studied in this paper, the loads were considered as constant impedances. However, the analysis must be extended to the other load characteristics such as; loads with constant power.
- (2) The effects of installing DG units on the methods performances are important and needs to be studied.

- (3) Performances of methods for the mesh networks should be studied.
- (4) Voltage sag source location in buses without any monitors can be studied using the results of an optimal monitoring program in transmission systems.

7. Conclusions

In this paper a new voltage sag source location method (CBM) based on DOC relay information was proposed. Location of the sag source has been identified by analyzing the variation of the magnitude of current positive-sequence component during sag and presag conditions and the sign of its phase-angle jump. This method showed 100% correct results under simulating faults in Brazilian power network. The performance of proposed method has also been verified by considering neutral grounding resistance effects. Proposed method used the current measurements only and it is very suitable in terms of accuracy, correct performance, computation time, the running speed and hardware costs.

In this paper, seven different methods have also been studied for voltage sag source relative location. These methods along with the proposed method have been applied under 80 different simulated voltage sags (symmetrical and asymmetrical) in large scale power network of the Brazilian. The results from different methods were studied and compared with each other.

In general, most methods demonstrated correct results for symmetrical faults (92.5% for both monitors). In asymmetrical faults, the correct results are approximately at 83% of cases. Results revealed that CBM, VSM and DR methods are the best with 100% accuracy in both symmetrical and asymmetrical faults. The results from this study can help utilities greatly in the operation and network planning.

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